

REACTJS SYLLABUS

OBJECTIVE: ReactJS is a declarative, efficient, and flexible JavaScript library for building reusable UI components. The objective of this training Session is to impart knowledge for the students and the Professionals who are starting their career in the React.

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1	Introduction to React
	Overview of React
	History and Evolution
	Core Concepts and Benefits
	Setting Up the Development Environment
2	Getting Started
	Installation and Setup
	Using Create React App
	Understanding project structure
	Hello World Application
	Rendering a simple component
3	ES6 Features
	Scope of variables
	Spread Operator
	Class and inheritance
	Destructuring
	Arrow Function
	Template Literals
	Map and Set
4	React Fundamentals
	Components
	Functional vs. Class Components - Class Components take again
	Component Lifecycle - Pending except componentDidMount
	Props and State
	Passing data with props
	Managing state within components
	State hooks (<i>useState</i>)
	Event Handling
	Handling user inputs - Pending for rendering validate
	Synthetic events - Pending
	Conditional Rendering
	Using <i>if</i> statements
	Ternary operators
	Short-circuit evaluation
5	Advanced React Concepts
	Hooks
	Introduction to Hooks
	<i>useEffect</i> for side effects
	<i>useContext</i> for context management
	Custom Hooks
	Context API
	Creating a context

	Using context in functional components
	Refs
	Creating and using refs with <code>useRef</code>
	Handling DOM elements
6	Component Design
	Component Composition
	Reusability and nesting components
	Higher-Order Components (HOCs) - Pending
	Render Props
	Controlled vs. Uncontrolled Components
7	Handling form inputs - Pending
	Managing component state vs. DOM state
	Routing
	React Router
	Setting up React Router
	Defining Routes
	Navigation with Links
	Route Parameters
	Nested Routes
8	Redirects and Route Guards
	State Management
	Local State Management
	Advanced state management with hooks
	Global State Management
	Introduction to Redux
9	Redux Toolkit
	Side Effects and Data Fetching
	Data Fetching
	Using <code>fetch</code> and <code>axios</code>
	Managing asynchronous operations
	Side Effects Management
	Handling side effects with <code>useEffect</code>
10	Data fetching patterns
	Ecosystem and Libraries
	Integrations
	Integrating with APIs
11	Working with third-party services
	Real-World Projects and Case Studies
	Project Development
	Building a full-featured application
	Applying concepts learned
	Case Studies
12	Analyzing real-world applications
	Discussing design choices and trade-offs
	Future Trends and Next Steps
	Continuing Education
	Recommended resources and communities
	Staying updated with React